

SMTP Guide for Thaghrach

Description

SMTP (Simple Mail Transfer Protocol) is the standard protocol used for **sending and relaying emails** across the Internet. It operates at the **Application Layer** and is responsible for transferring email messages from one mail server to another, or from an email client to a mail server.

SMTP is a **text-based, client-server** protocol that uses a simple command-response model. It is primarily designed for **sending** mail (not retrieving — that's done by POP3 or IMAP). Like HTTP and FTP, SMTP is human-readable, making it relatively easy to analyze. It typically runs on **TCP port 25**, with modern secure variants using port 587 (submission) or 465 (SMTPS).

Key Components of SMTP

1. SMTP Session Flow

A typical SMTP session follows this sequence:

- Client connects to the server
- Greeting from server
- Client identifies itself
- Sender and recipient information exchange
- Message transmission
- Session termination

2. Common SMTP Commands

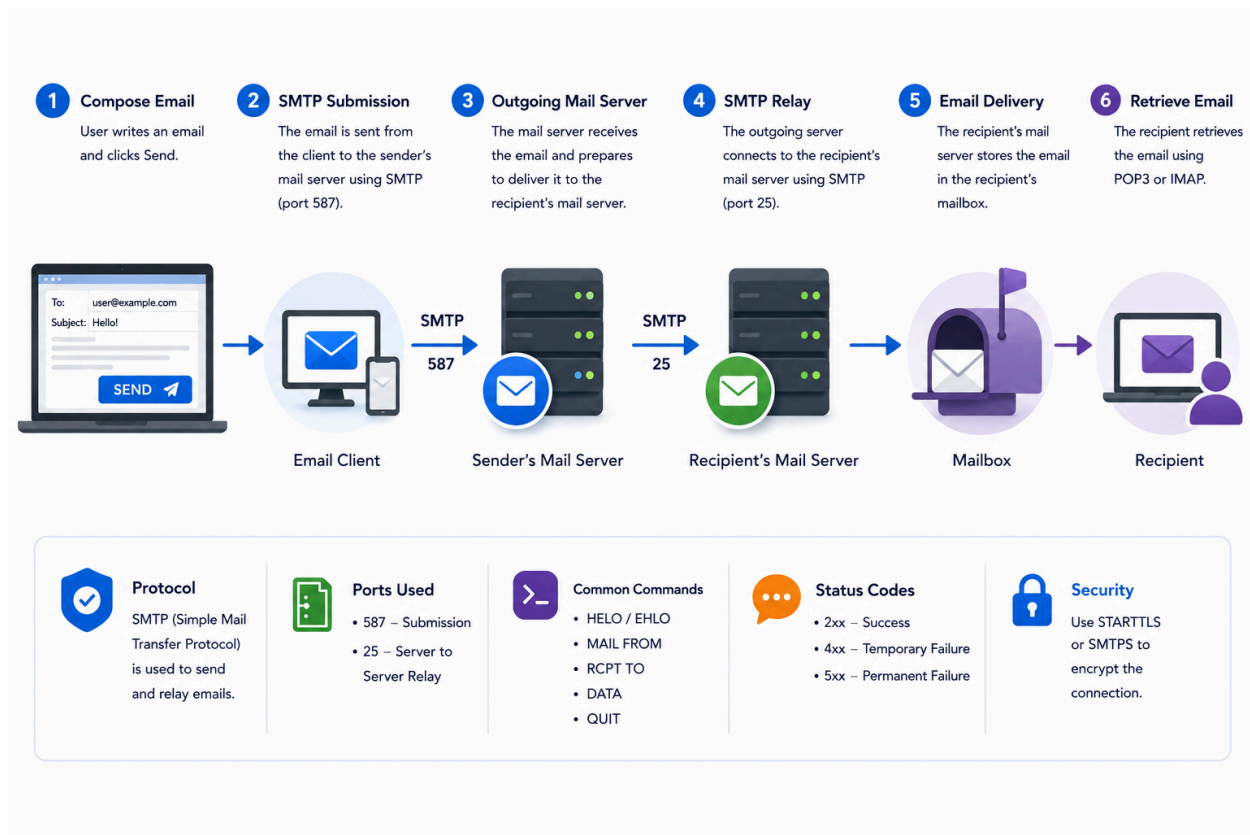
	COMMAND	PURPOSE	EXAMPLE
	 HELO / EHLO	Starts the SMTP conversation and identifies the client.	client: EHLO client.example.com server: 250 Hello mail.example.com
	 MAIL FROM:	Specifies the sender (email address).	client: MAIL FROM:<sender@example.com> server: 250 OK
	 RCPT TO:	Specifies the recipient (email address).	client: RCPT TO:<user@example.com> server: 250 OK
	 DATA	Indicates the start of the email content.	client: DATA server: 354 End data with <CR><LF>.<CR><LF>
	 . (End of Data)	A single period on a line by itself ends the message.	client: . server: 250 OK: message accepted
	 QUIT	Ends the SMTP session.	client: QUIT server: 221 Bye

3. Important Features

- **Email Headers:** From:, To:, Subject:, Date:, etc.
- **Message Body:** Plain text or HTML with possible attachments (via MIME)
- **Relaying:** Servers forward messages toward the final destination
- **Status Codes:** 3-digit reply codes (e.g., 250 OK, 550 No such user)

Visualizations

SMTP Email Transmission Flow:



What Is SMTP ?

<https://youtu.be/iUhDT3ZtWS0?si=iCUJn4E1vnIwKsTX>